

*Given a recording from a webcam,
can we reliably reconstruct 3D squat geometry and classify exercise quality?*

MediaPipe

We used Google MediaPipe for joint recognition.

Data Conversion

Due to training data being in the Kinect coordinate space we decided to transform all data to Mediapipe coordinate space before training.

Limited Datasets

We did not have access to that much training data ~180 files for training which impacted our results.

Data Augmentation Pipeline

Expanding limited training data to improve generalisation

X-Axis Mirror



Left↔right joint swap doubles dataset

Y-Axis Rotation



-10°, -5°, +5°, +10° rotations applied

XYZ Scaling



6 scale variants

Mirror + Rotate



Combined transforms: mirror each rotation

Mirror + Scale



Mirror applied to each scale variant

Gaussian Noise



$\sigma = 0.01$ per coordinate

Feature Engineering

Enriching raw 3D coordinates with biomechanically meaningful features

Base Coordinates

Hip-centred x, y, z per landmark

13 MediaPipe landmarks × 3 axes

= 39 raw features per frame

16 Inter-Joint Distances

Normalised Euclidean distances

between bone / limb pairs:

- Thigh, shin, torso, upper arm...

6 Joint Bend Angles

Cosine-based angle at each joint:

- Left & right knee
- Left & right elbow
- Left & right shoulder

Z-Axis Reconstruction Model

GRU / LSTM depth predictor

MODEL

Deep Learning: We trained a deep learning model to predict Z using our Mediapipe (x,y)

MEDIAPIPE

(x, y, z): MediaPipe also gives (x, y, z)

PIPELINE

Choice: We chose to use MediaPipes z-coordinates directly in our pipeline as the z predictor model where slow and gave similar results to MediaPipes coordinates

Start/Stop Model

Start/Stop model — isolating active exercise frames

Champion model result - MAE: 11.1178

HYPERPARAMETER	VALUES TESTED	WINNER
Optimiser	Adam vs SGD	Adam
Weight init	Xavier / Kaiming / Default	Default
Hidden size	64 / 128 / 256	128
Layer count	1 / 2 / 3 / 4	2
Sequence length	5 frames vs 30 frames	5
Dropout	0.1 / 0.2 / 0.3 / 0.4	0.1
Feature scaling	MinMax vs raw coordinates	MinMax

Good/Bad Classifier & Score Regressor

Pipeline dependency: both models consume sequences pre-cut by the A12 Start/Stop model, training on uncut data would pollute quality/score signal with rest frames.

GOOD / BAD CLASSIFIER

Model:	GoodBad_ClassifierV2
Input:	10 resampled frames × features
Features:	16 distances + 6 angles + coords
Augmentation:	Mirror, rotate, scale, noise
Segmentation:	Start/Stop model gates input
Output:	Good / Bad

SCORE REGRESSOR

Architecture	Description
ACNN	Attention + Conv
ARNN	Attention + RNN
Dense	Fully-connected baseline for comparison

Output: score 0.0–4.0
(0 = perfect form, 4 = poor form)

System Architecture

GitHub · main /
develop

GitHub Actions · CI/CD

lint · test · build · deploy on push

deploy

Browser

REACT + MEDIAPIPE

React + Vite + Tailwind CSS

MediaPipe: 33 landmarks in-browser

Sends 3D keypoints via HTTPS

Supabase JS, session storage

served by

Vercel

FRONTEND HOSTING

Global CDN edge network

API calls

Render

FASTAPI BACKEND

FastAPI · Python · Uvicorn

save session

Supabase

POSTGRESQL DATABASE

squat_keypoints table

Session frames + classifications

Written directly from browser

Docker

LOCAL DEV ONLY

docker compose up

Mirrors Vercel + Render split

load models

DagsHub / MLflow

MODEL REGISTRY

Prod / dev/ backup aliases

Full Pipeline Evaluation

End-to-end performance, after caching models

UNCHACHED - LATENCY BREAKDOWN (Finished recording → score)

MediaPipe Pose Det.	Network (Roundtrip)	Start/Stop	Good/Bad	Score Regressor	Total
~0ms	~1000ms	~1000ms	~5000ms	~4500ms	~11.5s

CACHED - LATENCY BREAKDOWN (Finished recording → score)

MediaPipe Pose Det.	Network (Roundtrip)	Start/Stop	Good/Bad	Score Regressor	Total
~0ms	~1000ms	~1000ms	~200ms	~200ms	~2.4s

Validation Problems

Domain Gap: Kinect → MediaPipe

Z-predictor is trained on Kinect-derived 3D data normalised to MediaPipe vocabulary, but at inference time it receives MediaPipe's own estimated landmarks. The coordinate systems differ subtly — this introduces a systematic bias that is hard to quantify without a paired ground-truth dataset.

Limited Dataset Size

Training corpus is a single Kinect dataset of squat exercises. Small N means cross-validation confidence intervals are wide. Augmentation helps but cannot substitute for true subject diversity (age, body type, footwear, camera angle).

Good Bad Classifier

Good vs. Ugly classification could not be trained reliably, the amount of data could not create a large enough testing dataset to get a complete valid result.

Pipeline consequential error

A consequential error in the pipeline, the start/stop impacts following models and if a ugly squat comes to scoring it can skew results.

Conclusions

①

Well implemented CI/CD improved efficiency

Using a well implemented CI/CD inside GitHub caused the project to move smoothly forward into new sprints.

②

Data methodology matters as much as model choice

Data augmentation and feature engineering improved results more than hyperparameter tuning.

③

Full session pipeline runs end-to-end in < 3 s from a browser

Record → segment → classify → score without specialist hardware

④

Free architecture causes disruptions

Using free variants like Vercel, Render creates disruptions in the development